

Requirements checklist & system architecture

Prepared for Dikie Kirk Orino, Pulido and Sanchez (adviser: Dr. Raphael Gonda). Revision 4 keeps the structure of the ChatGPT revision and adds a second review: restored code-level repository findings, infrastructure limits, buddy-finder safety, research ethics and tighter goal and evaluation rules. Rows shaded green changed in this revision. This is a design and verification plan, not a claim that the system has been built or validated.

Purpose and authority

A player-centred personal informatics web app: prepare, capture gameplay, review post-game evidence, receive peer feedback, and choose the next practice action. The five PPIF stages are Preparation, Collection, Integration, Reflection and Action.

The revision list transcribed in **THESISNOTES.pdf, p. 29** is the alignment basis. The original proposal-defense form was not supplied. Later notes retain post-game audio, integration of existing models, and individual and pair goals; these are kept distinct from the official-revision transcription.

ID	Required direction	Design response
R1	Player game performance	Players are primary users; no court-owner administration or health goals.
R2	Buddy for practice or competition	Opt-in matching by purpose, availability and compatible context; no public game listings.
R3	Pre-game, in-game and recovery data	Private participant check-in, recorded-game analysis and optional recovery log; missing entries explicit.
R4a	Peer coaching	Shared gameplay reports and timestamped partner notes in C1.
R4b	Video of practice or actual games	Upload-based analysis after play. Wall and ball-machine practice is logging-only in C1.
R5	Baseline metrics clarify goals	Per-metric baselines, explicit evidence rules, versioned resets and goal-linked comparisons.

How to use this specification

114 requirements across modules A to R. C1 = Capstone 1 target; C2 = Capstone 2 target; stretch = optional. IDs such as B03 are stable references. Tick an item only with test evidence. A quality-gated output that fails stays incomplete unless its scope change is recorded.

The 3-session baseline, 5-minute evidence minimum, 1 m line strip, release gates, retention period and processing budget are **team pilot choices**, not panel rules. Freeze them before final evaluation. The C1/C2 split is a planning assumption pending the adviser decision in REV11.

Corrections to carry into the thesis

Use REV01 to REV13 for manuscript issues; C1 and C2 stay reserved for delivery phases. Apply each correction consistently across Chapter 1, Chapter 3, diagrams, the Gantt chart and demo scenarios.

ID	Location	Issue	Resolution
REV01	1.4 S3; Objective 3; 1.7.5	Court-owner administration is included and excluded in different sections.	Replace S3 with buddy finding; remove queues, announcements and court-owner access management.
REV02	1.4 S4; Sprint 7; 3.5.2	Third-party coach and team roles conflict with peer coaching.	Use player and partner roles. Shared reports and timestamped peer notes are C1.
REV03	PPIF; 1.7.4; schedule	Real-time iPhone and live or semi-live audio wording survives.	Post-game processing only. Review cues come from an already processed session.
REV04	Sprint 7 vs 3.5.2	Access control and peer feedback have conflicting delivery phases.	Access control, sharing and peer notes are C1. Pair view, pair goals and audio are C2 planning targets.
REV05	1.7.2; Sprint 5	Provisional estimates are blurred with ratings.	Descriptive patterns are the default. Any inferred label needs an independent validation gate and stays non-official.
REV06	3.2.3; 3.4; 1.7.5	Fitting XGBoost on team features is model training.	No new model training in the current scope. Use released weights and documented rules; do not describe training as configuration.

Keep the claim proportional to the evidence

The system supports reflection and practice decisions. It does not establish that a movement pattern is good, diagnose technique from unsupported data, or prove that the app caused a player to improve.

Evidence, scope and delivery decisions

ID	Location	Issue	Resolution
REV07	1.7.1; 3.4	Shot accuracy, unforced errors and outcomes exceed guaranteed tracking evidence.	Exclude success and error labels and true 3D speed. Gate every metric and its downstream advice on validation and usable coverage.
REV08	R3; scope	The three capture phases are not operationally defined.	Create sessions before play; record participant capture times, timing provenance and recovery. Late entries are retrospective.
REV09	R5; goals	Historical comparisons do not define a baseline.	First 3 eligible sessions per player, metric, context and format. Persist membership and metric-specific aggregation; reset by creating a new version.
REV10	Chapter 2; notes p. 72	PB Vision and SwingVision are absent from the comparison.	Add dated primary-source comparisons. Frame baseline-linked goals, reflection and peer feedback as a contribution to evaluate, not established novelty.
REV11	Notes milestone; 3.5.2	A later note suggests all modules in C1; this plan assigns some to C2.	Keep the C1/C2 plan for scheduling, but obtain and record the adviser decision before committing dates. Do not describe it as approved.
REV12	3.4 tech stack	The paper says Pickleball-Analytics already uses ByteTrack. Its source calls plain per-frame YOLO predict, with no tracker.	State that persistent player IDs are the team's integration of Ultralytics ByteTrack, not an existing feature of the repository.
REV13	Chapter 3 ethics; data handling	The paper does not describe consent from people recorded in test footage or Data Privacy Act compliance.	Add a research-ethics and RA 10173 (Data Privacy Act) section: consent from every recorded person, retention periods and deletion rights.

Boundary decisions retained

- Recorded tournament footage may be analysed; tournament organisation is excluded.
- Wall and ball-machine practice can be logged, but neither receives C1 computer-vision analytics.
- Audio uses completed session results. It cannot delay or block publication of a successful core report.
- The app integrates released model weights. Rule configuration and evaluation are documented; training or fine-tuning is out of scope.

Requirements checklist

114 items in 18 modules, ordered along a player's session: set up, prepare, play, process, reflect, act. Green rows are new or changed in revision 4.

A. Accounts, profile and privacy

☐	A01	[C1]	Email sign-up and login via Supabase Auth; password reset works.
☐	A02	[C1]	Player profile: display name, dominant hand, years playing, usual format (singles, doubles), self-declared level.
☐	A03	[C1]	Record the uploader's recording-consent confirmation before first upload; store policy version and timestamp. Link a partner account only after that player accepts the invitation.
☐	A04	[C1]	Personal data is private to its player. Report access and peer notes follow explicit, revocable per-player share grants; verify database and storage access together.
☐	A05	[C1]	The uploader can delete a session and all video, tracks, results and audio. Show the effect on shared reports before deletion; linked players can remove their own personal entries and revoke shares.
☐	A06	[C2]	Player can export their data (CSV of metrics per session).

B. Preparation: goals and baseline

☐	B01	[C1]	Build a separate baseline for each player, metric, context and format from that metric's first 3 eligible sessions; session-level is_baseline flags are not used.
☐	B02	[C1]	Default eligibility: at least 5 minutes of metric-valid rally time and at least 80% tracking coverage of evaluable rally time. Shot shares also need at least 30 attributable contacts. Pilot targets, frozen after tuning and before evaluation.
☐	B03	[C1]	Before a baseline exists, the report shows raw values labelled "baseline in progress (n of 3)", with no deltas.
☐	B04	[C1]	Baseline stored per metric with its range, aggregated by metric type as the data dictionary defines.
☐	B05	[C1]	Baseline stays fixed until the player deliberately resets it; each reset is logged with a reason and creates a new baseline version.
☐	B06	[C1]	After a reset, the next 3 eligible sessions form the new baseline; existing goals stay linked to the baseline version they were set against.
☐	B07	[C1]	Numeric goals use only validated comparable metrics: kitchen-line presence, zone shares, mean rally duration, eligible shot shares, and mean occupied area once its own gate passes. Draft a focus before a baseline; activate a numeric goal after it is ready.
☐	B08	[C1]	Each goal stores a target, direction, context, format, baseline version, start date and deadline. The target must lie in the chosen direction from the baseline; no metric is better merely because it is higher.
☐	B09	[C1]	Starter goal suggestions come from the player's stated focus; later suggestions use evidence from sessions after the baseline.
☐	B10	[C1]	Update goal status on each eligible session and deadline check (awaiting evidence, on track, behind, target reached once, achieved, deadline missed) using the goal-progress policy. Missing or invalid data never counts as success.
☐	B11	[C1]	Completing a goal prompts the player to set the next one.
☐	B12	[C2]	Pair goals link two consenting players and one individual goal per player; each member sees the shared goal progress only. Both member goals must be achieved to complete the pair goal.
☐	B13	[C1]	Glossary tooltips for pickleball terms (kitchen, dink, third shot, reset).

C. Pre-game check-in

☐	C01	[C1]	Create a session before play. Each participant may record warm-up, sleep hours, readiness and session focus before the actual start time; attach video later.
☐	C02	[C1]	Compare captured_at with the recorded actual session start, not upload time. Entries after play begins are retrospective; an unknown start time is labelled timing unverified.
☐	C03	[C1]	Show each player their own check-in in the private report; missing values stay missing, never default to zero.
☐	C04	[C1]	No medical or health scoring; stored only as context for performance.

D. Video capture and upload

☐	D01	[C1]	Recording guide screen: tripod height, behind-baseline angle, full court visible, 30 fps minimum, landscape.
☐	D02	[C1]	Resumable chunked upload of MP4 and MOV. The size limit follows the storage plan chosen under P07; a 10-minute 1080p phone clip is typically several hundred MB, so show the limit before upload and offer trimming.
☐	D03	[C1]	Upload attaches to the session created at check-in, or creates one; form: date, context, format, partner (optional), notes.
☐	D04	[C1]	Player selects themselves on the earliest usable frame. Partner identity requires an accepted link; unlinked participants keep anonymous roles. Track IDs are local to a processing run, not permanent identities.
☐	D05	[C1]	Court corner confirmation: auto-detected keypoints shown, player can drag to correct.
☐	D06	[C1]	Validate original file metadata before conversion: minimum 720p and 30 fps. Reject insufficient court visibility, unreadable footage or unsupported media with a reason; upscaling does not make footage eligible.
☐	D07	[C1]	Processing status visible: queued, processing with percent, done, failed with reason.
☐	D08	[C1]	Wall practice and ball-machine sessions are logged with check-in, recovery and tags only; no CV analysis in C1, since the pipeline needs a full visible court.

E. Computer vision pipeline

☐	E01	[C1]	FFmpeg and OpenCV with aspect ratio preserved. Keep source timestamps and a source-to-analysis frame map; keep the original frame rate when supported, otherwise document and validate resampling.
☐	E02	[C1]	Detect and validate 12 court keypoints, with manual correction available; recompute homography on camera drift and store calibration and court-side segments.
☐	E03	[C1]	YOLOv8 detects players; Ultralytics ByteTrack (added by the team, not present in Pickleball-Analytics) provides track continuity, subject to identity checks and corrections.
☐	E04	[C1]	Store time-bounded track-to-participant mappings and an append-only correction history. A correction invalidates affected metrics, goals and audio until they are recomputed.
☐	E05	[C1]	Player position projected to court metres. The ball stays in image coordinates except at bounces, where it touches the court plane.
☐	E06	[C1]	Store raw and smoothed ball tracks with observed and interpolated flags. Interpolate only short bounded gaps (initial limit 0.10 s, frozen before evaluation). Interpolation alone never establishes a contact or bounce.
☐	E07	[C1]	Segment rallies using ball presence, motion and temporal context; record complete versus truncated rallies and rally confidence.
☐	E08	[C1]	Contact events: frames where the ball changes direction near a player.
☐	E09	[C1]	Detect candidate bounces from trajectory evidence and review against labels. A vertical image-direction change alone is not proof of a bounce; the volley rule needs an observable no-bounce interval.
☐	E10	[stretch]	MediaPipe pose landmarks for ready-position and split-step cues.
☐	E11	[C1]	Each event stores confidence and evidence provenance. Metric quality also uses valid coverage and dependency checks; a detector score is not a metric accuracy probability.
☐	E12	[C1]	Raw per-frame tracks saved as a file (Parquet or JSON) in storage, not as DB rows.

F. Metrics and shot classification

☐	F01	[C1]	Kitchen-line presence: share of valid tracked rally time in the 1 m strip immediately behind the non-volley line on the player's own side. Time inside the kitchen is a separate zone.
☐	F02	[C1]	Report non-overlapping kitchen, transition, baseline and outside-court shares; use the spatial dictionary and show missing coverage separately.
☐	F03	[C1]	Report observed distance and mean occupied area (convex hull) per eligible rally. These describe movement; they do not measure defensive effectiveness.
☐	F04	[C1]	Complete-rally count, mean and maximum complete-rally duration, and tempo per observed minute, as defined in the dictionary.
☐	F05	[C1]	Player heatmap in court coordinates.
☐	F06	[C1]	Shot type per contact from documented rules: serve, volley, dink, drive, lob, or unclassified when evidence is missing or weak.

CHECKLIST

☐	F07	[C1]	Labelling guide fixes precedence (serve, then volley, then dink, drive, lob); arc and speed are image-space proxies validated against labels, not true 3D values.
☐	F08	[C1]	Shot distribution per session, with unclassified shown as its own share.
☐	F09	[C2]	Rally-end proximity: whether a rally ends within 2 subsequent contacts after a shot, when that window is fully observed. Not shot success, accuracy or a winner/error label.
☐	F10	[C1]	Metrics validated against manually labelled clips; accuracy recorded per metric.
☐	F11	[C1]	Hide unvalidated or failed metrics. Session values lacking coverage are unavailable or clearly low confidence, and excluded from baselines, goals, recommendations and audio.
☐	F12	[C1]	For each metric show valid rally seconds over evaluable rally seconds, observed versus interpolated share, and contact count where relevant. Session length is shown separately.

G. Post-game report and dashboard

☐	G01	[C1]	Session report: KPIs, heatmap, shot distribution, rally summary, check-in, goal progress.
☐	G02	[C1]	Compare each eligible KPI with its baseline version and previous eligible session in the same context and format; otherwise show why comparison is unavailable.
☐	G03	[C1]	Filter trends by context, format and metric-definition version. Never silently mix values produced with incompatible definitions.
☐	G04	[C1]	Show progress for each active goal toward its target. Do not rank raw units against normalised values; show up to 2 largest progress increases or decreases only when comparable goal evidence exists.
☐	G05	[C1]	Annotated clip links: tap a metric event to jump to that moment in the video.
☐	G06	[C1]	Default view shows 4 to 6 KPIs; details in expandable sections.
☐	G07	[C1]	Reflection prompt after each report: what went well, what to change, free text.

H. Post-game recovery log

☐	H01	[C1]	Optional post-session recovery log per participant: exertion 1 to 10, soreness 1 to 5, cooldown done, capture time. A skipped log is visibly missing.
☐	H02	[C1]	Shown beside performance trends as context only; no injury or medical output.

I. Recommendations

☐	I01	[C1]	Rule engine maps the gap between the active goal and the current value to focus areas and drills; more movement, longer rallies or more dinks are never treated as better by default.
☐	I02	[C1]	Each recommendation identifies its goal, triggering value, processing run and reviewed drill source. Never use an invalid or stale metric.
☐	I03	[C1]	Next-session target generated from the active goal.
☐	I04	[C1]	Player can accept, dismiss or mark a recommendation done; status stored.
☐	I05	[C2]	Effect check compares later eligible sessions for the targeted metric; describe association and goal progress without claiming the app caused improvement.

J. Audio feedback and retrieval

☐	J01	[C2]	Retrieval knowledge base holds reviewed tactics, drills, cue templates and source metadata. C1 recommendations use a small curated drill table and deterministic rules, with no pgvector dependency.
☐	J02	[C2]	Post-match spoken summary (60 to 90 s) generated per processed match; toggle per upload.
☐	J03	[C2]	Review cues: short event-based cues (under 12 words, one action) from the processed session, playable between drills.
☐	J04	[C2]	Block unsupported numbers, low-quality evidence, medical advice and officiating claims. Compare generated figures to structured metrics, log rejections, and fall back to an approved template.
☐	J05	[C2]	Store audio and transcript per player and processing run; neither reaches a partner merely because the gameplay report was shared.
☐	J06	[C2]	Feedback log: event, retrieved sources, generated text, timestamp.

K. Play patterns and optional labels

☐	K01	[C1]	Descriptive play-pattern feedback by default, for example "drive share rose from X to Y" across comparable eligible sessions; no inferred level or style label by default.
☐	K02	[C1]	Inferred level and style labels stay disabled unless evaluated against a predefined rubric and independent experienced raters on held-out players. Self-ratings are context, not ground truth.
☐	K03	[C1]	An enabled label is provisional, lists its evidence and thresholds, and needs at least 3 eligible matches. Validation is a separate release gate, not implied by match count.

L. Gameplay context

☐	L01	[C1]	Required context tag on upload: practice, casual, tournament, leveling, drill.
☐	L02	[C1]	Format tag: singles, doubles, wall practice, ball machine.
☐	L03	[C1]	All trends and baselines filterable by context and format.

M. Peer coaching and pairs

☐	M01	[C1]	Player can link a partner by invite code; either side can unlink.
☐	M02	[C1]	A per-player, per-session share grant controls that player's metrics, clips and heatmap; private forms, reflections, recommendations, goals and audio stay private.
☐	M03	[C1]	Partner leaves timestamped notes on a shared session; the player sees them in the report (basic peer feedback, R4a).
☐	M04	[C2]	Pair view combines separately consented player reports; neither account can share the other player's personal records.

N. Buddy finder

☐	N01	[C1]	Opt-in buddy profile: area, availability, format, purpose and self-declared level. A validated provisional level is optional; new players stay eligible without one.
☐	N02	[C1]	Rank buddies by availability, purpose and format, with level closeness only when both have usable level data. Omit missing criteria, renormalise the weights and show the matching reasons.
☐	N03	[C1]	Send and accept buddy requests; accepted buddies can be linked as partners.
☐	N04	[C1]	No venue booking, no payments, no public game listings.
☐	N05	[C1]	Safety: area stored at city or municipality level only; block, report and hide-profile controls; buddy finder open to players who confirm they are 18 or older.

O. Engagement

☐	O01	[stretch]	Optional consistency challenges follow the player's chosen goals, such as logging 3 sessions in a week. Never reward more kitchen time, distance or a particular shot share by default.
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P. Non-functional requirements

☐	P01	[C1]	Pilot performance target: a 10-minute 1080p clip processes in under 30 minutes on a named worker configuration. Record processing and queue time separately; a test target, not a measured result.
☐	P02	[C1]	Claim jobs atomically with a lease and attempt token; one active analysis per session. Expired leases can be retried, but stale workers cannot publish. Publish complete versioned results atomically.
☐	P03	[C1]	Failed jobs retry once, then report the failing stage.
☐	P04	[C1]	Videos, tracks, heatmaps, clips and audio are private. Authorise each signed URL request; URLs last at most 5 minutes. Revocation blocks new URLs immediately.
☐	P05	[C1]	Responsive web app usable on a phone at the court.
☐	P06	[C1]	Record model and checkpoint identifiers, code revision, metric definitions, thresholds, calibration and preprocessing settings per processing run.
☐	P07	[C1]	Choose the storage plan before Sprint 1 ends. Supabase Free caps files at 50 MB and total storage at 1 GB and pauses idle projects, so full-length video needs Pro, another object store, or trimming to rallies.
☐	P08	[C1]	Retention: delete the raw video a set period after successful processing (suggest 30 days) unless the player keeps it; tracks, metrics and clips stay. State the period in the consent screen.

Q. Testing and evaluation

☐	Q01	[C1]	Test video set covering singles and doubles, two lighting conditions, two camera heights, recorded with consent from every visible player.
☐	Q02	[C1]	Split by original recording before tuning; keep alternate cuts of one match together. Freeze held-out footage and pass criteria before inspection. Further tuning needs a new untouched final set.
☐	Q03	[C1]	Pilot labels: at least 200 contacts and 20 complete rallies, with at least 20 held-out reference contacts per shot class. A class below that is reported as insufficient evidence, not scored. Double-label a subset, report agreement, adjudicate.
☐	Q04	[C1]	Evaluate the full pipeline: event precision and recall, shot class metrics including abstentions, rally boundaries, zone-time error in percentage points, identity switches and coverage, per condition with sample counts.
☐	Q05	[C1]	Apply the pilot release gates. A failed dependency disables downstream metrics and advice and does not count as a completed feature; record any scope reduction in the manuscript.
☐	Q06	[C1]	Reliability log: failed uploads, processing time on the named GPU, incomplete tracking, failed-job recovery.
☐	Q07	[C2]	User evaluation with players and partners tests baseline interpretation, evidence-based practice priority, sharing and notes. Specify recruitment, sample rationale and scoring beforehand; usability survey is secondary.
☐	Q08	[C2]	Sample data for the demo scenarios in 3.5.3, updated for the revised scope.

R. Paper and deliverables

☐	R01	[C1]	Resolve REV01 to REV13 across problems, objectives, scope, diagrams, schedule and evaluation; update the paper to the verified implementation.
☐	R02	[C1]	Distinguish panel revisions from team design choices (baseline sessions, evidence minima, spatial bands, accuracy and processing targets); document rationale and versions.
☐	R03	[C1]	Add PB Vision and SwingVision to the related-systems review using dated primary evidence. Test the proposed contribution; do not claim uniqueness without comparison.
☐	R04	[C1]	Code and checkpoint attribution plus a dependency register with exact versions, availability and reuse terms.
☐	R05	[C1]	Metric definitions appendix matching the data dictionary.
☐	R06	[C2]	Final manuscript reflects the built system and test results.
☐	R07	[C1]	Obtain research-ethics clearance through the DLSU process and write the RA 10173 data-privacy note before collecting evaluation footage or running user tests.

Incomplete evidence is a normal state

Use clear states such as waiting for video, insufficient coverage, baseline in progress, processing failed and metric unavailable. Never show a zero where the system has no reliable measurement. Buddy matching still works for a new player with no analysed matches.

Metric dictionary: spatial metrics and coverage

Coordinates. x runs 0 to 6.096 m across the court width; y is distance from the net toward the player's own baseline, 0 to 6.7056 m. Non-volley line $L = 2.1336$ m. Dimensions follow USA Pickleball [S1]. Store court-side intervals so end changes do not reverse zone meaning.

Position proxy. Bottom-centre of the player box projected to court coordinates. It approximates floor position; it is not a measured foot landmark. Player gaps of at most 0.5 s may be interpolated between valid endpoints; longer gaps are excluded, and gaps never bridge identity or side changes.

Terminology. "Coverage" always means tracking coverage (how much of the rally was measured). Court area a player moves over is called **occupied area**.

Metric / unit	Session definition	Baseline aggregation
Kitchen-line presence / %	100 x valid seconds in the strip $L \leq y < L+1.0$, within sidelines, divided by valid position seconds. Time inside the kitchen is excluded.	Pool strip seconds / valid position seconds across the 3 eligible sessions.
Zone shares / %	Kitchen: $0 \leq y < L$. Transition: $L \leq y < 5.2$. Baseline: $5.2 \leq y \leq 6.7056$. Outside: valid positions outside the own-side rectangle. Mutually exclusive, sum to 100%.	Pool zone seconds / valid position seconds. The line strip is a subset of the transition zone.
Observed distance / m	Sum of displacement between consecutive valid, filtered positions within one continuous rally segment. Excluded gaps are not bridged. More distance is not better.	Mean of eligible session totals, shown descriptively with observed minutes. Not a numeric goal in C1.
Mean occupied area / m^2	Convex-hull area of valid positions within each sufficiently covered complete rally, then the mean across those rallies. An occupancy proxy, not defensive effectiveness.	Pool eligible rally areas / eligible rally count. Goal-eligible only after its own release gate passes.
Player heatmap	Occupancy map in court coordinates, normalised by valid tracked time, side-normalised, with legend.	Visual comparison only; no numeric baseline or goal.
Tracking coverage / %	100 x metric-valid rally seconds / evaluable rally seconds. Report both durations and the interpolated share. Session length is shown separately.	Quality metadata, not a performance baseline.

The 1 m strip, 5.2 m boundary, gap limits and evidence thresholds are configurable team choices; save their definition version. Kitchen occupancy alone never establishes a fault; no officiating output is produced.

Metric dictionary: rallies, shots and context

Metric / phase	Definition and quality rules	Baseline / use
Rally count / in-game	Detected complete rallies. Truncated rallies are listed separately; evaluate count precision and recall, not boundary timing alone.	Mean session count, descriptive, with elapsed duration beside it.
Mean / max rally duration / in-game	Mean and maximum of complete-rally durations. Clips cut at the start or end are excluded.	Mean: pooled rally seconds / count. Max: largest across baseline rallies, labelled baseline maximum.
Rally tempo / in-game	60 x complete rally count / observed session-window seconds, including pauses between rallies but excluding off-camera periods.	Pooled count / pooled observed window, x 60.
Shot distribution / in-game	Class contacts / all attributable detected contacts, with unclassified in the denominator; shares sum to 100%. Unattributed contacts reported separately.	Pooled class counts / attributable contacts. Needs validated contact detection and enough observed ball coverage.
Ball motion proxy / C2	Pixels per second in a fixed analysis frame, with frame size and calibration segment saved. Not km/h, arc height or camera-independent speed.	Diagnostic only; no cross-camera goal or baseline. Homography is not used for airborne ball height [S2].
Rally-end proximity / C2	Did the rally end within 2 subsequent contacts after this shot? Unknown if that window was not observed.	Event descriptor only. Never success, accuracy, winner or error.
Check-in / pre-game	Warm-up yes/no; sleep hours; readiness 1 to 5; optional focus goal. Capture time compared with actual session start.	Private context; no medical score or CV baseline.
Recovery / post-game	Perceived exertion 1 to 10; soreness 1 to 5; cooldown yes/no; capture time. Missing is not zero.	Private context; no health goal, diagnosis or injury prediction.
Play-pattern summary / derived	Describes comparable validated shot shares or positioning changes, with values and evidence window.	Default output. Inferred skill or style stays disabled without independent validation.

Not reported: official in/out calls, kitchen faults, spin, paddle angle, unforced errors or shot success rates. A change in image position is not sufficient evidence for any of them.

A reproducible improvement loop

Baseline eligibility and versioning

For each player, metric, context, format and metric-definition version, take the first 3 eligible sessions after enrolment or reset, ordered by played_at with session ID as tie-breaker. Default eligibility: 5 minutes of metric-valid rally time, 80% or more tracking coverage, passed metric validation, and for shot shares 30 or more attributable contacts. These minimums are pilot choices, not evidence of statistical sufficiency.

Baseline membership is independent per metric; one session may qualify for positioning but not shots. Store numerators, denominators, run IDs and per-session values. The session min to max range is descriptive, not a confidence interval.

A reset creates a new version from the next 3 eligible sessions and records its reason. Existing goals keep the prior baseline. If a source session is deleted, reprocessed or fails quality checks, mark the baseline and dependent goals stale and recompute or ask for a deliberate replacement; never silently rewrite historical results.

Goal progress: one scale, clear states

Let B be the goal's frozen baseline value, T the target, and X the latest eligible value in the same context and format. Require T different from B and a direction consistent with T minus B. Progress $P = (X - B) / (T - B)$. Show the raw change $X - B$ in metric units and P as target progress; cap the progress bar visually at 0 to 100% but keep the raw value.

Elapsed fraction $E = \text{time since activation} / \text{time from activation to deadline}$, capped at 0 to 1. Status rule:

- **Target reached once** if $P \geq 1$ on the latest eligible session only.
- **Achieved** if $P \geq 1$ on 2 consecutive eligible sessions. One good session can be noise; requiring a repeat keeps goals honest without extra statistics.
- Before the deadline: **on track** if $P \geq E$, otherwise **behind**.
- After the deadline: **deadline missed** if not achieved (a reached-once result is noted).
- No eligible evidence since activation: **awaiting evidence**.

This schedule comparison is a simple planning rule, not a prediction. Before a baseline exists, only a stated focus or draft goal is allowed. Achievement records its supporting runs; a later decline shows in trends without erasing it.

Recommendations and pair goals

C1 uses reviewed drills and deterministic rules keyed to the active goal, showing the triggering metric, observed value, target and drill source, without implying a causal diagnosis. C2 pair goals reference two individual goals; members share goal progress only, while check-ins and recovery stay private. Comparative cards use target progress for valid goals only; with one valid goal, show one card.

Candidate repositories and evidence register

Revision 3 removed the code-level findings. They are restored here because they were observed directly: README files and core modules were read on the default branch on 29 September 2026 (court_detection.py, player_tracker.py and ball_tracker.py for Pickleball-Analytics). Findings still need a pinned commit and an inference test before any dependency commitment.

Candidate	Observed on 29 Sep 2026	Role and required next check
kpp91302/ Pickleball-Analytics	YOLO 12-keypoint court model with RANSAC homography to a bird's-eye plane; players projected by box bottom-centre; ball box with NaN interpolation; heatmaps, kitchen presence, rally tempo; Tkinter GUI. Players use plain per-frame YOLO predict (no tracker). Ball confidence threshold 0.15. Requires exactly 12 ordered keypoints. No LICENSE file.	Backbone candidate. With no licence, reuse needs written permission from the author; ask now and keep a fallback. Remove the GUI for headless use; add ByteTrack (REV12).
AndrewDettor/ TrackNet-Pickleball	TrackNetV2 fine-tuned for pickleball; TensorFlow; weights hosted on Google Drive; GPU needed for practical speed.	Ball tracker, inference only. Download and hash the weights, confirm terms, and compare against the YOLO ball box on the same held-out footage.
Surabhi/ Pickle_ball_tracking	MIT licence. Ultralytics detection with FilterPy Kalman smoothing, bounce detection and in/out logic.	Borrow Kalman smoothing and bounce logic with attribution; validate bounces independently. Do not ship in/out calls.
Ryan Tolone: Pickleball Vision	Write-up of calibration and match analytics; code availability not confirmed.	Reference only.
Roboflow: pickleball-d etection-1oqlw	Hosted dataset and model.	Fallback detector or extra validation data; verify terms and keep it separate from final test footage.
Arduino paddle tracker	Hardware project.	Excluded: custom hardware is out of scope.
bradautomates/claude -video; Vinod Polinati rally-splitter article	Dev tool and article.	Optional labelling aids; manually verify anything used as ground truth.

Dependency record required before Sprint 2

For each selected package or checkpoint save: source URL, commit, weight identifier and hash, licence or permission evidence, installed versions, preprocessing contract, inputs, outputs, hardware, measured latency and test result. If a candidate cannot be used, pick an alternative within the no-training scope or mark the capability as unproven.

Shot rules and processing sequence

Single-label convention: serve, then volley, then dink, drive, lob, otherwise unclassified. This is an application convention, not a claim that the concepts never overlap; explain it in dashboard help and the labelling guide. Changing it changes shot shares and needs a new metric version.

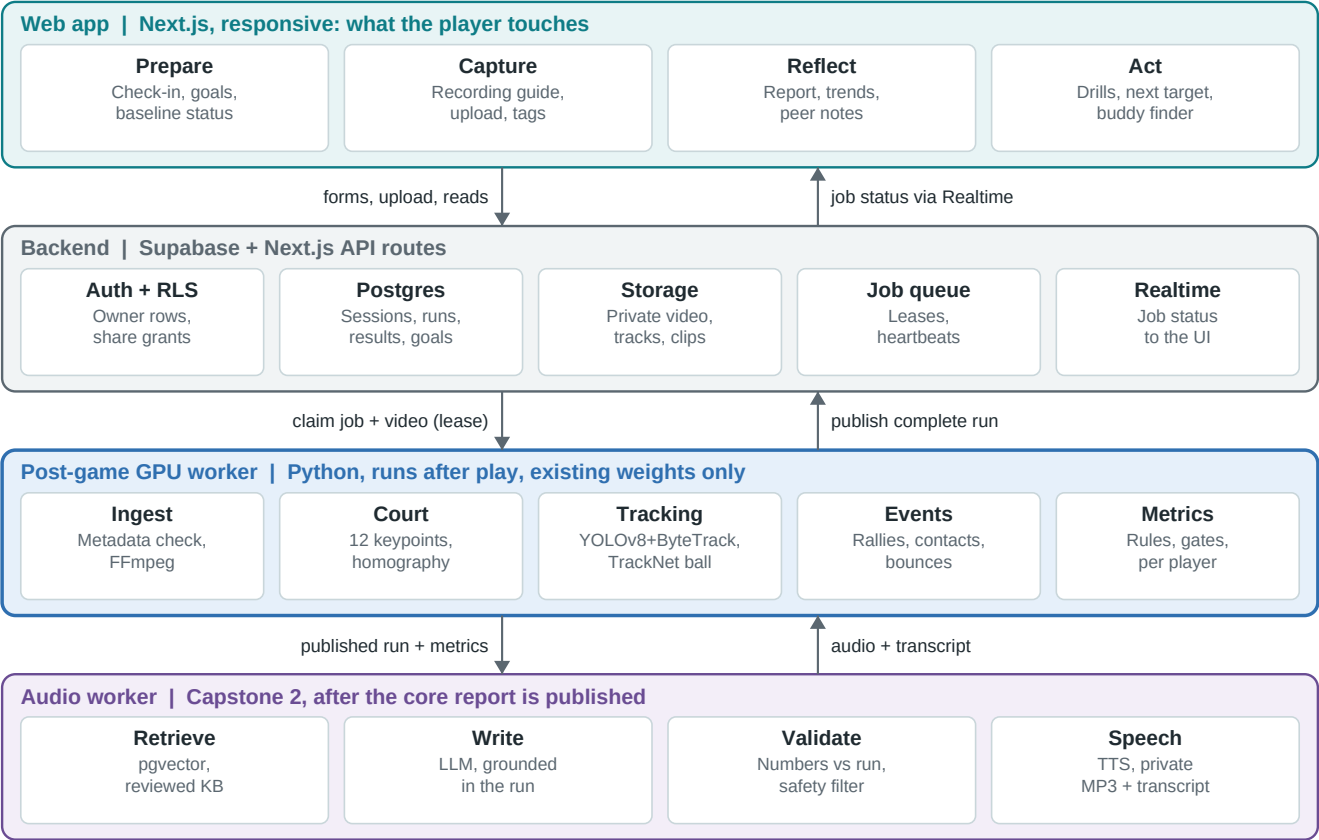
Class	Minimum evidence
Serve	A confidently observed rally start and contact by a player behind the baseline, with compatible sequence context. The first contact in a clipped video is not automatically a serve.
Volley	Contact after an opponent hit, with a continuously observed interval that supports no intervening bounce. A missed bounce detection is not evidence of no bounce.
Dink / drive / lob	Calibrated context plus validated image-space trajectory proxies. Dink: near-kitchen contact and observed opponent-kitchen landing. Drive: fast, flat observed path. Lob: high image-space path and observed deep landing. Abstain if evidence is missing or rules conflict.
Unclassified	Weak or missing ball data, ambiguous contact, truncated context or conflicting rules. Context is re-established from sufficient observed events, not one isolated bounce.

Per-run pipeline

1. **Validate and ingest.** Check original metadata and sampled court visibility; normalise only as needed and keep a source timestamp map.
2. **Calibrate and identify.** Detect and correct court keypoints; store camera and side intervals. Track players and apply time-bounded account mappings.
3. **Track and infer events.** Track the ball, bound gaps, then detect rallies, contacts and bounces with observed or interpolated provenance.
4. **Classify and measure.** Apply versioned rules; compute per-player metrics, sufficient statistics, quality gates and coverage.
5. **Publish core results.** Write artifacts and rows under a run ID, then atomically publish the complete set and notify the web app.
6. **Update personal views.** Update eligible baselines, goals and recommendations; mark stale dependants.
7. **Generate audio separately (C2).** Retrieve reviewed knowledge plus current metrics, validate the text, create speech, publish per player. Failure leaves the core report usable.

System architecture

One post-game analysis path. The browser never calls the GPU worker; it writes a job through the backend, the worker claims it with a lease, publishes a complete run, and Realtime tells the web app it is ready. Job notifications are status updates, not live video analysis.



PicklePro architecture: four lanes, post-game only. Revision 3 reduced this to four plain boxes; the component view is restored so each lane shows what it owns.

Component	Planned technology	Responsibility / PPIF stage
Web UI	Next.js, Tailwind, Recharts; hosted on Vercel	Forms, upload, goals, reports, peer workflows / preparation, reflection, action.
Backend	Supabase Auth, Postgres, Storage, Realtime; Next.js routes	Identity, policies, records, private artifacts, job queue / integration.
GPU worker	Python; YOLOv8, ByteTrack, TrackNet, OpenCV, FFmpeg, FilterPy	Existing-model inference, events and metrics / collection, integration.
Audio (C2)	pgvector, an LLM API, a TTS API	Reviewed retrieval, checked summaries, speech / reflection, action.

Vercel cannot host the GPU worker, since long video jobs exceed serverless limits. Choose and benchmark the host (lab machine, Modal or RunPod) before the performance test. Names here are a proposed stack, not a working-integration claim.

Data model: sessions, participants and runs

Logical fields and relationships. UUID keys, foreign keys, uniqueness and ownership constraints are enforced. A session is a recording context; a participant is a person in it; a track is a temporary model output.

Entity	Key fields	Invariant / purpose
profiles	id, display_name, hand, years_playing, usual_format, self_level, is_adult_confirmed	One row per user. Adult confirmation gates the buddy finder (N05).
consent_records	id, player_id, policy_version, accepted_at, scope, retention_days	Timestamped consent, separate from report-sharing grants; shows the retention period (P08).
sessions	id, uploader_id, actual_start_at, actual_end_at, context, format, video_path, video_expires_at, status, active_run_id	One uploader; nullable video for logging-only sessions. No global is_baseline flag or single partner_id.
session_participants	id, session_id, player_id nullable, role, team_side, link_status	Account link accepted by that person; anonymous partner and opponent roles; one linked row per player per session.
checkins	participant_id, captured_at, readiness, sleep_hours, warmup, focus_goal_id, timing_status	Private. timing_status: pre-game, retrospective or unverified, derived against actual_start_at.
recovery_logs	participant_id, captured_at, rpe, soreness, cooldown	Private. Nullable values mean missing, never zero.
analysis_runs	id, session_id, status, input_hash, code_revision, model_versions, config, metric_version, started_at, completed_at	Immutable provenance: preprocessing, thresholds, calibration and quality summary.
jobs	id, run_id, kind, stage, progress, attempts, lease_token, lease_expires_at, heartbeat_at, error	One active core job per session; audio jobs have their own kind and target player.
track_assignments	id, run_id, track_id, participant_id, start_s, end_s, court_side	Time intervals handle switches, occlusion and end changes; conflicting assignments rejected.
track_corrections	id, assignment_id, old_mapping, new_mapping, actor_id, at, reason	Append-only. Affected results regenerate before showing as current.

Data model: versioned results and decisions

Entity	Key fields	Invariant / purpose
rallies	id, run_id, start_s, end_s, complete, confidence	Complete versus truncated; source-video timestamps.
contacts	id, rally_id, participant_id nullable, track_id, t_s, class, proxy_px_s, evidence, confidence	Unattributed contacts kept with null participant. No speed_kmh or success field.
metric_results	run_id, participant_id, metric_key, definition_version, value, numerator, denominator, valid_s, eligible_s, count, quality_status	Unique per run, participant and metric; sufficient statistics allow correct pooling. Values may be unavailable.
baselines	id, player_id, context, format, metric_key, definition_version, version, aggregation, value, min, max, status, created_at, reset_reason	Value is not always a mean. Status: building, ready, stale, retired. Versions retained.
baseline_members	baseline_id, session_id, run_id, participant_id, metric_key	Exactly 3 contributions for a ready baseline; deletion invalidates dependent use.
goals	id, player_id, metric_key, target, direction, context, format, baseline_id, start_at, due_at, status, reached_once_run_id, achieved_run_ids	Numeric goal needs a ready matching baseline; supporting runs recorded.
reflections	id, participant_id, went_well, change_next, captured_at	Private; not exposed by a report share.
drills / rules	id, title, content, source, version; goal_metric, condition, drill_id	C1 curated knowledge and deterministic rules; no vector search.
recommendations	id, participant_id, run_id, goal_id, metric_key, trigger_value, drill_id, rule_version, status	Accept, dismiss or done; invalid metrics cannot trigger advice.
artifacts	id, run_id, participant_id nullable, kind, private_path, state	Tracks, heatmaps, clips, audio; per-run paths stop stale retries overwriting current outputs.

Share gameplay without sharing personal logs

Entity	Key fields	Purpose
partners / buddy_requests	id, from_player_id, to_player_id, status, created_at	Invite, accept and unlink lifecycle; no automatic association from video.
buddy_profiles	player_id, city, availability, time_zone, format, purpose, self_level, opted_in, hidden	City-level area only. Missing level does not block matching.
blocks / reports	id, reporter_id, target_id, kind, reason, created_at	Block hides both players from each other's suggestions; reports go to a moderator queue.
report_shares	id, session_id, subject_player_id, recipient_id, scope, granted_at, revoked_at	Grantor is the subject player; covers their metrics and gameplay artifacts only.
peer_notes	id, share_id, author_id, t_s, body, created_at, updated_at	Author needs an active share; timestamp points into the authorised video.
pair_goals (C2)	id, player_a, player_b, goal_a_id, goal_b_id, accepted_a, accepted_b, status	Consented progress only, not private logs.
kb_chunks (C2)	id, source_url, reviewed_by, version, text, embedding	Reviewed knowledge provenance and retrieval trace.
audio_feedback (C2)	id, run_id, participant_id, kind, text, audio_path, sources, validation_status	Private per-player output; numbers checked against the run.

Access rules that must pass tests

- A player reads and changes their own personal records. Owning a recording does not expose another participant's private logs.
- A recipient with an active share reads only that player's gameplay report, heatmap and authorised clips. Recordings may visually include others; the consent flow says so.
- The recipient creates notes and edits or deletes only their own; the subject may delete any note on their report. Revocation hides the report and notes from the recipient; notes stay with the subject.
- Unlinking revokes active shares between the two accounts. New signed URLs stop immediately; issued ones expire within 5 minutes.
- Blocking a buddy removes both from each other's suggestions and cancels pending requests.
- Session deletion runs as a tracked job; dependent baselines and goals are marked stale rather than keeping misleading values.

Pilot release gates and failure behaviour

Proposed pilot acceptance choices, not measured results. Freeze them after tuning and before opening the final evaluation set. Report sample counts and per-condition results; pilot success does not establish accuracy for all courts or players.

Output / gate	Pilot criterion	Failure behaviour
Rally detection	Event precision and recall each 0.90 or higher; 90% of matched complete rallies have both boundaries within 1 s. Fix a one-to-one matching rule first.	Suppress rally totals and rally-window metrics until corrected.
Position / zone time	Mean absolute zone-share error of 10 percentage points or less per player-session on annotated windows; 80% position coverage for an eligible session.	No baseline or goal update from failed sessions; show coverage and reason.
Contacts and shots	Contact precision and recall each 0.80 or higher within a 0.2 s window. Shot macro F1 of 0.70 or higher over classes with at least 20 held-out references; unclassified 30% or less. Classes below 20 references are reported as insufficient evidence.	Disable shot goals and dependent advice if the contact or class gate fails.
Player identity	Report switches per tracked minute and wrong-player attribution. Published clips have no unresolved swaps.	Require correction or omit affected player metrics.
Distance / occupied area	Compare on the same labelled windows; predeclare a tolerable error before exposing them as validated.	Hide until passed; no baseline or goal.
Runtime / recovery	10-minute 1080p clip under 30 minutes on named hardware; retry after worker loss, duplicate-request deduplication and stale-worker rejection shown.	Clear stage failure; no duplicate or partial results.
Permissions	All negative access tests pass for two partners plus an unrelated account, including blocks and revoked grants.	Block release of sharing until fixed.
Inferred skill / style	Disabled by default. Predefine an independent-rater rubric, sample plan, agreement measure and threshold before enabling.	Descriptive patterns and self-declared level only.

Why the per-class minimum: 200 pilot contacts split across five classes leaves serves and lobs with very few examples. A macro F1 computed on a handful of lobs is not meaningful, so small classes are reported honestly instead of averaged in.

C2 user study: recruit the intended player group and practice partners, justify the sample size, and score tasks against a rubric (interpret baseline, identify an evidence-supported practice priority, control sharing, use a peer note). Improvement trends are observational, not causal proof.

Failure-safe jobs and reproducible publication

Surface	Responsibility
Authenticated client + policies	Profiles, participant forms, goals, reflections, buddy profiles and requests, blocks, shares, peer notes. Derived results are read-only.
POST /api/sessions/:id/process	Check uploader authority, input and configuration; create a versioned run and job with an idempotency key; return the current job on duplicates.
GET /api/buddies/suggest	Opt-in, adult-confirmed, unblocked candidates with ranking reasons; no private session data.
POST /api/sessions/:id/audio (C2)	Check the target player and published core run; enqueue independent audio work.
Authorised artifact endpoint	Check participant or share scope, then issue a signed URL with at most 5-minute expiry.
Worker poller (internal)	Atomically claim eligible jobs with a lease token. No public worker endpoint; service credentials stay server-side.
Retention sweeper (scheduled)	Deletes raw video past video_expires_at unless the player chose to keep it; logs each deletion.

State and concurrency contract

Session states: draft, awaiting video, queued, processing, ready, failed, deleting, deleted. Logging-only sessions never enter the CV queue. Audio has its own queued, processing, ready and failed states.

The queue gives retryable processing, not exactly-once execution. Claim in a transaction, heartbeat a time-limited lease, and compare the lease token before publishing; a replaced worker cannot publish. Retry a failed or lost job once, then show the failed stage and a user-triggered retry.

Write artifacts and rows under run-specific keys and publish by switching active_run_id only after all required outputs validate. When mappings, inputs or rules change, create a new run and mark dependent baselines, goals, recommendations and audio stale; never mix old and new metric rows.

Test evidence to capture

Input hash, timestamps, checkpoint and rule versions, hardware, stage durations, peak memory, attempts, error category and run ID. Test interrupted uploads, worker loss, duplicate enqueue, deletion during processing and unauthorised artifact access.

Build order and scope guardrails

Keep the seven-sprint structure and prove the CV path early. The C2 split is a planning assumption subject to REV11.

Phase	Focus	Exit evidence
1	Accounts, participants, private storage, upload, context tags, simple check-in; choose storage plan and GPU host; request Pickleball-Analytics permission.	Session exists before play and video attaches later; two accounts cannot read each other's private data; P07 decision recorded.
2	Headless CV worker, calibration, player and ball tracking, rally segmentation.	Tracks and rallies for at least 3 recordings; dependency record complete.
3	Contacts, bounces, shot rules, spatial metrics, annotation and tuning; test the 5-minute baseline minimum on real footage.	Pilot feasibility report; thresholds and evaluation protocol frozen; final test set unopened.
4	Reports, baselines, goals, reflection, recovery.	A player sees baseline progress, valid comparisons and explicit missing-data states.
5	Reviewed drill rules, recommendations, descriptive play patterns.	Every recommendation links to valid evidence and a reviewed drill; labels stay gated.
6	Partner invitations, sharing, notes, buddy finder with safety controls.	Partner comments on an authorised report; revocation, block and access tests pass.
Buffer	C1 acceptance and scope review.	Requirement-by-requirement evidence and recorded scope decisions.
7	Integration, runtime and final held-out evaluation.	End-to-end report plus frozen-set results.
C2	Audio and RAG, pair view and pair goals, user evaluation, manuscript.	Grounded audio, consented pair workflows, study results and limitations written up.

Out of scope

Court booking, payments, owner or organiser dashboards, open-play queues, public listings and announcements; tournament, league or bracket organisation; official ratings or officiating; health goals, injury risk or diagnosis; live analysis or audio; training or fine-tuning any model; custom hardware and wearables. C1 excludes CV for wall and ball-machine practice. Recorded tournament footage stays an allowed input.

Decisions still open

- **C1/C2 split** with Dr. Gonda (REV11).
- **Buddy finder in C1**: R2 keeps it, but an earlier note said finding a partner is not a system feature. Confirm.
- **Storage plan** (P07) and **GPU host**; both have cost implications for a student team.
- **Pickleball-Analytics permission** from its author, or a fallback backbone.
- **Ethics clearance timing** (R07), since evaluation footage cannot be collected before it.

What changed and what remains to verify

Revision 4 (this version): second review of the ChatGPT revision

- Restored the code-level repository findings from the 29 September inspection (no tracker and no licence file in Pickleball-Analytics, ball threshold 0.15, 12-keypoint requirement, MIT licence for Surabhi) and added REV12 for the ByteTrack misattribution in section 3.4.
- Added infrastructure reality: Supabase Free caps files at 50 MB [S3], so the 2 GB upload in revision 3 was not achievable on that plan (P07, D02), plus a raw-video retention rule (P08) and sweeper.
- Added buddy-finder safety (N05): city-level area, block and report, adults only, with matching tables and API rules.
- Added research ethics and RA 10173 data-privacy compliance (REV13, R07) and consent for everyone in test footage (Q01).
- Goals now need the target met on 2 consecutive eligible sessions before "achieved"; one session shows "target reached once".
- Shot evaluation needs at least 20 held-out references per class before a class counts toward macro F1 (Q03, gates).
- Fixed terminology: "coverage" now only means tracking coverage; court area is "occupied area" (B07, F03). B06 now says eligible, and F04 now matches the dictionary.
- Restored the four-lane component architecture diagram in place of the four plain boxes.
- Added a decisions-still-open list; checklist grows from 110 to 114 items.

Revision 3: ChatGPT editorial revision (retained)

- Stable requirement IDs; the kitchen-line strip separated from kitchen occupancy; exclusive zones with explicit endpoints.
- Target-based goal progress, baseline membership and versions, late check-in timing, per-player ownership, anonymous participants.
- Time-bounded track identities, granular sharing, run provenance, lease-protected publication, independent audio processing.
- Final-test isolation, percentage-point errors, abstentions and failure cascades; all thresholds marked as pilot choices.

Sources

Supplied: ThesisPaper.pdf (July 13 proposal), THESISNOTES.pdf (official revisions p. 29; later notes pp. 30 to 31; repositories p. 33; related systems p. 72), and the revision 3 PDF. The original defense form was not supplied.

[S1] USA Pickleball, court construction and official dimensions. <https://usapickleball.org/construction/> Accessed 29 Sep 2026.

[S2] OpenCV, Basic concepts of the homography (court-plane mapping versus airborne ball). https://docs.opencv.org/4.x/d9/dab/tutorial_homography.html

[S3] Supabase docs, Storage file limits: Free projects cannot exceed 50 MB per file; Pro and up allow up to 500 GB. <https://supabase.com/docs/guides/storage/uploads/file-limits> Accessed 29 Sep 2026. Confirm total storage and project-pausing rules on <https://supabase.com/pricing> before choosing a plan.

[S4] Republic Act No. 10173, Data Privacy Act of 2012 (Philippines), and its implementing rules via the National Privacy Commission.

Repository findings come from reading the repositories directly on 29 September 2026; product comparisons for Chapter 2 still need their own dated primary sources.